

Score Normalization by Dynamic Time Warping

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Abstract— Multi-expert approach is a well-known paradigm to support complex decisions and several decision fusion techniques have been considered. In this field score normalization is a fundamental step to allow high performances and hence several techniques have been proposed so far. This paper addresses the problem of score normalization and presents a new technique based on Dynamic Time Warping. The experimental results, carried out in the field of pattern classification, show the superiority of the new technique with respect to other approaches in the literature, based on MIN-MAX, z-score and characteristics functions.

Keywords; *Human-Computer Interaction, Multi-Expert Systems, Decision Fusion, Dynamic Time Warping.*

I. INTRODUCTION

In the last years, multi-expert approach has been considered an effective paradigm to support difficult decisions. When multi-expert approach is considered, many experts are used for a single problem, and their decisions are combined to obtain the final decision [1,2].

On the basis of the kind of decisions combined, the methods for expert combination can be divided into three categories: Measurement-level combination methods combine the scores provided by the individual experts. In this case each score is considered the degree of confidence of the expert for the decision provided; Ranked-level combination methods combine ranked lists of decisions provided by the individual experts, ordered according to their degree of confidence; In Abstract-level combination methods combine the top-decision decisions provided by the experts [2,3].

Among the three categories, the combination of experts at the measurement-level is expected to be the most effective, since it uses all information available. Unfortunately, the performance of a multi-expert system based on measurement-level combination, strongly depends on the effectiveness of the strategy used for normalizing the scores provided by the individual experts. Indeed, the output produced by each individual expert can depend on several aspects and hence different experts produce outputs in different ranges. In addition, also when the range is the same for two different experts, the meaning of the same score produced by the two experts can be different, when – for instance – it is obtained by different similarity/dissimilarity measures or membership functions. Therefore, incomparability of the output scores for different experts is a major problem in expert combination. In order to deal with this problem, the scores need to be

normalized [4,5]. In the past, several approaches have been considered for output normalization based on output range resizing, statistical adaptation, characteristic functions [6,7,8].

This paper uses Dynamic Time Warping (DTW) for output normalization and shows its performance in the context of multi-classifier systems using sum- and product- rules. The organization of the paper is as follows: Section 2 presents the state-of-the-art techniques for output normalization in the context of pattern classification. The technique based on DTW is introduced in Section 3. Section 4 presents the experimental results, carried out in the field of handwritten numeral recognition. The conclusions of the present work and some directions for further research are highlighted in Section 5.

II. SCORE NORMALIZATION TECHNIQUES

Let A be a measurement-level classifier that associates input patterns to the classes of the set

$$\Omega = \{\omega_1, \omega_2, \dots, \omega_i, \dots, \omega_M\}.$$

In this case, given an unknown pattern x , A produces a set of output values

$$\begin{aligned} A(x) &= (a_1, a_2, \dots, a_i, \dots, a_M) = \\ &= (A(x, \omega_1), A(x, \omega_2), \dots, A(x, \omega_i), \dots, A(x, \omega_M)) \end{aligned}$$

where each matching score $a_i = A(x, \omega_i)$ is a confidence value supporting the evidence that x belongs to ω_i .

In the ideal case a_i is the a-posteriori probability of ω_i given the input pattern x . Sometimes, a_i is just an approximation of the a-posteriori probability. In many practical cases, however, a_i is the result of a matching between the input pattern x and the class ω_i , computed using similarity or dissimilarity measures. Therefore, scales and distributions of the output values provided by different classifiers are generally very diverse, since they strongly depend on many parameters, such as features used for pattern matching, similarity/dissimilarity measures, etc.

Hence, the problem of output normalization is very significant when results from different classifiers have to be combined, in order to exploit potential of classifier combination. Standard techniques proposed in literature for output normalization are the min-max technique [6], the z-score technique [6] and the normalization technique based on the characteristic function [7,8].

2.1 Min-Max technique

Min-max normalization [6] is the simplest normalization technique that is best-suited for the cases where the bounds of the scores produced by a classifier are known:

- $S_{\min}$ = minimum of the scores;
- $S_{\max}$ = maximum of the scores.

In this case, given a set of matching scores a_i , $i=1,2,\dots,M$, the set of normalized scores is given by:

$$a'_i = \frac{a_i - S_{\min}}{S_{\max} - S_{\min}} \quad i=1,2,\dots,M. \quad (1)$$

Clearly, when the matching score are not bounded, $S_{\min}$ and $S_{\max}$ can be estimated.

2.2 Z-score technique

Z-score normalization is the most commonly used score normalization technique [6]. This technique works well if prior knowledge about the average score and the score variations of the classifier is available. If there is no a priori knowledge about the nature of the matching algorithm, the mean and the standard deviation of the scores should be estimated from a given set of matching scores. Given a set of matching scores a_i , $i=1,2,\dots,M$, the set of normalized scores is given by:

$$a'_i = \frac{a_i - \mu}{\sigma} \quad i=1,2,\dots,M \quad (2)$$

where μ is the arithmetic mean and σ is the standard deviation of the given data set.

2.3 Characteristic function technique

Output normalization can be achieved by aligning the accumulated recognition rate, that is a continuous, monotone growing function over classifier output [7,8]. For this purpose the correct recognition function of a classifier is first computed, that counts, for each likelihood value, the number of correctly recognized samples $n_{\text{correct}}(\cdot)$ [7,8].

In this case, given a set of matching scores a_i , $i=1,2,\dots,M$, the set of normalized scores is given by:

$$a'_i = a'_i + \text{charf}(a_i) \quad (3)$$

where the *characteristic function* $\text{charf}(\cdot)$ is defined as [7]:

$$\text{charf}(a_i) = a_{\max} \cdot r_i - a_i \quad (4)$$

being

- $a_{\max}$ the maximum possible output of the classifier;
- r_i the partially accumulated recognition rate (where N is the number of patterns):

$$r_i = \frac{\sum_{j=1}^i n_{\text{correct}}(j)}{N} \quad (5)$$

III. SCORE NORMALIZATION BY DTW

This Section presents in detail the new technique for score normalization. It uses Dynamic Time Warping (DTW) to match the cumulative recognition rate of a classifier against a standard curve, that is the same for all classifiers. More precisely, in this paper, the accumulated Gaussian probability function has been considered as standard curve ($\mu = 0$, $\sigma = 1$). Now, let

$$R = (r_1, r_2, \dots, r_L) \quad (6)$$

be the L-point discrete version of the accumulated recognition function and

$$G = (g_1, g_2, \dots, g_L) \quad (7)$$

be the L-point discrete version of the accumulated probability function of the standard Gaussian distribution. A warping function between R and G is any sequence of couples of indexes identifying points of R and G to be joined [9]:

$$W(R, G) = c_1, c_2, \dots, c_T, \quad (8)$$

where $c_t = (i_t, j_t)$, being T, i_t, j_t integers, $1 \leq t \leq T$, $1 \leq i_t \leq L$, $1 \leq j_t \leq L$.

Now, if a distance measure $d(\cdot, \cdot)$ is considered between points of R and G ,

$$d(c_t) = d(r_{i_t}, g_{j_t}), \quad (9)$$

we can associate to $W(R, G)$ the dissimilarity measure

$$D_{W(R, G)} = \sum_{t=1}^T d(c_t). \quad (10)$$

It is worth noting that although several distance measures $d(\cdot, \cdot)$ can be considered depending on the applications, in this paper the Euclidean distance has been used.

The elastic matching procedure detects the optimal warping function $W^*(R, G) = c^*_1, c^*_2, \dots, c^*_{T^*}$ which satisfies the conditions of

- monotonicity,
- continuity,
- boundary conditions,

and for which it results [9]:

$$D_{W^*(R,G)} = \min_{W(R,G)} D_{W(R,G)} . \quad (11)$$

Now let be $W^*(R, G)$ the optimal warping function for the matching between R and G , given a matching score a_i , $i=1,2,\dots,n$, its normalized value is given by:

$$a'_i = \frac{1}{card(G_i)} \cdot \sum_{g_q \in G_i} g_q \quad (12)$$

where G_i is the set of samples of G associated to the sample r_i of the accumulated recognition function, i.e.:

$$G_i = \{ g_q \mid (q,i) \in W^*(R, G) \}. \quad (13)$$

In other words, the normalized value of a score a_i is given by the average value of the points of the accumulated Gaussian function which corresponds to the sample r_i , being r_i the sample of the accumulated recognition function that corresponds to a_i . Thus, Dynamic Time Warping is used to map the value of the accumulated recognition function r_i (corresponding to the confidence value a_i) on the standard curve (the accumulated Gaussian function). The average of the values on the standard curve that corresponds to r_i is considered as the normalized value a'_i corresponding to a_i .

IV. EXPERIMENTAL RESULTS

In order to test the proposed technique the multi-classifier system in Figure 1 has been considered. The input pattern x is fed to K individual classifiers in parallel. Each classifier A_i provides its response $A_i(x)$, $k=1,2,\dots,K$. The responses obtained by all the classifiers are then combined, according to a suitable combination strategy, to obtain the final results $E(x) = E(A_1(x), A_2(x), \dots, A_K(x))$ [1,2].

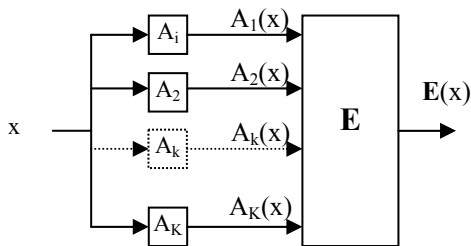


Figure 1. Multiclassifier System

The system has been designed for handwritten digits recognition and four different classifiers have been considered ($K=4$), whose description is beyond the aims of this paper and the interested reader can find more details in refs. [10, 11]. The classifiers use:

- A_1 : Histogram-based classifier;
- A_2 : Intersection-based classifier;
- A_3 : Zoning-based classifier ;
- A_4 : Pattern Matching classifier.

Output combination has been performed by sum-rule and product-rule [12]. Let x an unknown input pattern and a_i^k the confidence value confidence value, produced by the classifier A_k , supporting the fact that x belongs to ω_i ;

- when sum-rule is considered, the following measure is considered:

$$B_i = \sum_{s=1}^K a_i^{ts} , \quad i=1,2,\dots,M \quad (14)$$

with $a'_i{}^k$ being the normalized version of a_i^k ;

- when product-rule is considered, the following measure is considered:

$$B_i = \prod_{s=1}^K a_i^{ts} , \quad i=1,2,\dots,M \quad (15)$$

with $a'_i{}^k$ being the normalized version of a_i^k .

In both cases the pattern x is classified to the class ω_j if and only if

$$B_j = \max_{(i=1,2,\dots,K)} B_i \quad (16)$$

The CEDAR database has been considered [13]: 18467 patterns for learning (BR directory) and 2189 patterns for the test (BS directory).

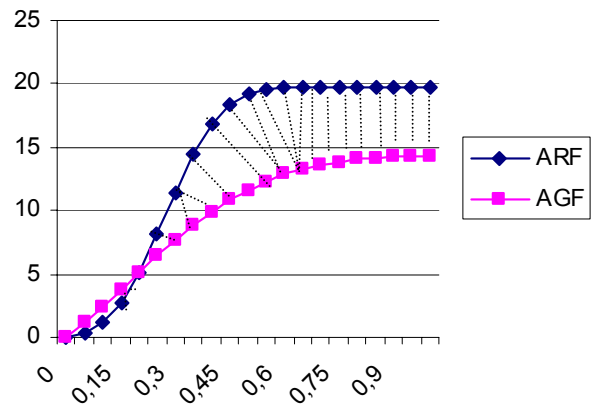


Figure 2. Example of Normalization by DTW

Figure 2 shows the results of DTW between the Accumulated Recognition Functions (ARF) of the histogram-based classifier and the Accumulated Gaussian probability Function (AGF). In this case the Euclidean distance is used for matching. Tables 1 reports the performance of the multi-expert system when the sum-rule and the product-rule are used. In both cases DTW normalization procedure allows the achievement of the best performance. Specifically, when the sum-rule is used, DTW normalization outperforms MIN-MAX, Z-Score and Characteristic Function techniques respectively of 1.6%, 10.9% and 5.2%. When the product rule is considered, the DTW normalization technique and the Characteristic Function technique provide the same results. Furthermore, they outperform MIN-MAX and Z-Score techniques of 2.4% and 56.2%, respectively.

Table 1. Recognition Rate

Normalization	SUM RULE	PRODUCT RULE
MIN-MAX	79,2	78,4
Z-Score	72,6	51,4
Characteristic Function	76,5	80,3
DTW-based	80,5	80,3

V. CONCLUSION

In this paper a new approach for output normalization of measurement-level multi-expert system is presented, based on Dynamic Time Warping, and its effectiveness is experimentally evaluated in the context of sum-based and product-based classifier combination. Some experimental results, carried out in the field of handwritten digit recognition, demonstrate the superiority of the proposed technique with respect to other approaches in the literature, based on MIN-MAX, Z-Score and Characteristic Functions.

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